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MEASUREMENT OF HIGH CURRENTS USING A SHUNT

Abstract

A current measurement device arranged to measure a main current passing therethrough and includes a first element which is electrically conductive and which includes a shunt and a second element which is electrically conductive. Each end of the first element is fastened to a separate end of the second element by an electrically conductive connection such that the main current is split into a first current flowing in the first element and a second current flowing in the second element, the shunt thus making it possible to measure the first current, which is representative of the main current.

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Background/Summary

[0001] The invention relates to current measurement devices.

BACKGROUND

[0002] Shunts are conventionally used in electricity meters to measure the current supplied by a distribution network to an installation. The measurements of the current are then used to evaluate the power and electrical energy consumed by the installation.

[0003] FIG. 1 shows a prior-art shunt 1. This shunt 1 is intended to be integrated in an electricity meter that measures currents of less than 100 A.

[0004] The shunt 1 comprises a main segment 2 in the form of a rectangular plate, and a first terminal 3a and a second terminal 3b located at the ends of the main segment 2. Copper plates 4a, 4b are fastened to the ends of the shunt 1 on either side thereof. Wires (not shown) are connected to each of the terminals 3a, 3b and allow the voltage across the terminals of the shunt 1 to be measured. The current flowing in the shunt 1 is evaluated on the basis of these voltage measurements.

[0005] By way of example, the shunt 1 has a length $l_{\text{sub.1}}$ of 5 mm, a height $h_{\text{sub.1}}$ of 8 mm and a thickness $e_{\text{sub.1}}$ of 1.79 mm, and therefore a cross section along a plane perpendicular to its length $l_{\text{sub.1}}$ of 14.3 mm^{sup.2}. The impedance of the shunt 1 at 20° C. (almost entirely resistive at the frequencies under consideration) is then 150 $\mu\Omega$.

[0006] The two copper plates 4a, 4b each have a length of 12.5 mm, a height of 8 mm, a thickness of 1.79 mm and an impedance at 20° C. of 15 $\mu\Omega$. The total impedance of the two copper plates at 20° C. is 30 $\mu\Omega$.

[0007] An electrically conductive part having a total length of 30 mm and a total impedance at 20° C. of 180 $\mu\Omega$ is thus obtained.

[0008] This shunt 1 was designed for a maximum current to be of 90 A. The power dissipated inside the measured electricity meter by the shunt 1 at this current is 1.46 W, which is acceptable.

[0009] The aim is to design a current measurement device for a meter that measures high currents (200 A or 320 A, for example). At first glance, it would seem logical to use a shunt such as that in FIG. 1.

[0010] However, if a current of 200 A passes through this shunt 1, the power dissipated at this high current is 7.2 W. At a current of 320 A, the power obtained is 18.44 W.

[0011] These powers are too high for electricity meters which, in their standard design, cannot dissipate those powers effectively.

[0012] To reduce the dissipated power, the height and/or thickness of the shunt can be increased.

[0013] For example, at a shunt height of 100 mm, the dissipated power obtained is 1.47 W. With a constant height, a thickness of 23 mm would be required in order to obtain a comparable result. However, these dimensions are too large for the shunt to be integrated in a standard electricity meter. Furthermore, these solutions lead to significant additional costs.

[0014] It is also possible to reduce the length of the shunt. However, even at a length of 1 mm, which is the absolute minimum length, the dissipated power obtained is 6.44 W, which is again too high.

Object

[0015] The object of the invention is to measure high currents using a measurement shunt while limiting the dissipated power, with very good measurement accuracy and lower costs.

SUMMARY

[0016] With a view to achieving this object, a current measurement device is proposed, which is arranged for measuring a main current passing therethrough and comprises: [0017] a first element which is electrically conductive and which includes a shunt; [0018] a second element which is

electrically conductive; each end of the first element being fastened to a separate end of the second element by an electrically conductive connection such that the main current is split into a first current flowing in the first element and a second current flowing in the second element, the shunt thus making it possible to measure the first current, which is representative of the main current.

[0019] The materials used to make the first element (and thus the shunt) and the second element can be selected such that the impedance of the first element is much higher than that of the second element. The power dissipated in the shunt when the first current passes therethrough and, more generally, the power dissipated by the current measurement device when the main current passes therethrough are then significantly lower than if the entire main current passed through the shunt.

[0020] The first current flowing in the shunt is a resemblance of the main current, and the accuracy of the measurement is very high. If necessary, the temperature drift differences in the materials used can be compensated for.

[0021] The cost of the current measurement device is low. This device uses a “classic” shunt which can be used for low currents, and a second element which can be a very simple and inexpensive part or segment of a part.

[0022] In addition, a current measurement device as described above is proposed, comprising a part that comprises a side in which there is formed a notch that defines a recessed portion extending along a segment of a length of the part, and a non-recessed portion extending along said segment of the length of the part, the second element being said non-recessed portion, and the first element extending in the notch so as to leave a non-electrically conductive space between the first element and the second element.

[0023] In addition, a current measurement device as described above is proposed, in which each end of the first element comprises a first cavity formed in its body, and each end of the second element comprises a second cavity formed in its body, the first element and the second element having their ends nested in such a way that a bottom of the first cavity in each end of the first element abuts a bottom of the second cavity in a separate end of the second element.

[0024] In addition, a current measurement device as described above is proposed, in which the second element is made of at least one material comprising a copper alloy.

[0025] In addition, a current measurement device as described above is proposed, in which the first element comprises the shunt and two plates, which are fastened to the shunt and positioned on either side of the shunt.

[0026] In addition, a current measurement device as described above is proposed, in which the shunt is made of at least one material comprising Manganin.

[0027] In addition, a current measurement device as described above is proposed, in which each electrically conductive connection is established by a spot-welding operation.

[0028] In addition, a current measurement device as described above is proposed, in which an electrical resistance of the first element is between 5 times and 100 times an electrical resistance of the second element.

[0029] In addition, an electrical apparatus including a current measurement device as described above is proposed, the electrical apparatus comprising a processing unit arranged to: [0030] measure a voltage across the terminals of the shunt; [0031] evaluate the main current on the basis of said voltage.

[0032] In addition, an electrical apparatus as described above is proposed, further comprising a temperature sensor, the processing unit being further arranged to acquire temperature measurements produced by the temperature sensor and to compensate, depending on said temperature measurements, for a temperature drift difference of an electrical resistance of the first element and an electrical resistance of the second element.

[0033] In addition, an electrical apparatus as described above is proposed, the electrical apparatus being an electricity meter.

[0034] In addition, an electrical apparatus as described above is proposed, the electrical apparatus

being arranged to measure a main current greater than 100 A using the current measurement device. [0035] In addition, a current measurement method is proposed, which is carried out in the processing unit of the electrical apparatus as described above and comprises the steps of: [0036] measuring a voltage across the terminals of the shunt; [0037] acquiring the temperature measurements produced by the temperature sensor; [0038] determining, depending on the temperature, a value of a compensation parameter for compensating for the temperature drift difference of the electrical resistance of the first element and of the electrical resistance of the second element; [0039] correcting the voltage across the terminals of the shunt using said value of said compensation parameter; [0040] evaluating the main current on the basis of the corrected voltage.

[0041] In addition, a computer program is proposed, comprising instructions which cause the processing unit of the electrical apparatus as described above to execute the steps of the current measurement method as described above.

[0042] In addition, a computer-readable storage medium is proposed, on which the computer program as described above is stored.

[0043] The invention will be better understood in the light of the following description of a particular, non-limiting embodiment of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0044] Reference will be made to the accompanying drawings, in which:

[0045] FIG. 1 is a perspective view of a prior-art shunt;

[0046] FIG. 2 shows a meter in which a current measurement device is integrated;

[0047] FIG. 3 is a perspective view of the current measurement device;

[0048] FIG. 4 is a side view of the current measurement device.

DETAILED DESCRIPTION

[0049] With reference to FIG. 2, the electricity meter **10** is intended for measuring the electrical energy consumption of an installation **11**. This electrical energy is supplied to the installation **11** by a distribution network **12**. The distribution network **12** comprises a phase line **14** and a neutral line **15**. In this case, the meter **10** is a single-phase meter.

[0050] The meter **10** comprises an upstream phase terminal P connected to the phase line **14** of the network **12**, and an upstream neutral terminal N connected to the neutral line **15** of the network **12**. The meter **10** also comprises a downstream phase terminal P' and a downstream neutral terminal N'.

[0051] In this case, the term “upstream” means on the distribution network **12** side, and the term “downstream” means on the installation **11** side.

[0052] The downstream phase terminal P' and the downstream neutral terminal N' of the meter **10** are connected to the installation **11**, optionally via respective switches integrated in a circuit breaker located outside the meter **10** and not shown here.

[0053] The meter **10** also comprises a phase conductor **16**, which is connected to the phase line **14** of the distribution network **12** via the upstream phase terminal P and interconnects the upstream phase terminal P and the downstream phase terminal P'. The meter **10** also comprises a neutral conductor **17**, which is connected to the neutral line **15** of the distribution network **12** via the upstream neutral N and terminal terminal N and the interconnects the upstream neutral downstream neutral terminal N'.

[0054] In this case, the meter **10** also comprises a cut-off member **18** comprising a switch assembled on the phase conductor **16**. The cut-off member **18** is used in particular for remotely interrupting or re-establishing the supply of power to the installation **11**, e.g. in the event of the

subscription being cancelled or the subscription contract not being complied with.

[0055] The meter **10** also comprises metrological components.

[0056] These metrological components comprise a voltage sensor (not shown) which measures the voltage applied by the network **12** to the terminals of the installation **11** (and of the meter **10**).

[0057] The metrological components also comprise a current measurement device **20**, which is assembled on the phase conductor **16** upstream of the cut-off member **18**. The current measurement device **20** makes it possible to measure the current I that flows on the phase line **14** and on the phase conductor **16** and that is supplied to the installation **11** by the distribution network **12**.

[0058] Here, the meter **10** includes at least one temperature sensor (two in this case). These sensors comprise a temperature sensor **21**, which is positioned close to the current measurement device **20**. The temperature sensor **21** produces temperature measurements that are representative of the temperature of the current measurement device **20** and its immediate surroundings. The role of this temperature sensor **21** will be described later.

[0059] The meter **10** further comprises a processing unit **22** (electronic and software). The processing unit **22** comprises at least one processing component **23**, which is, for example, a “general-purpose” processor, a processor specialising in signal processing (digital signal processor (DSP)), a microcontroller, a programmable logic circuit, such as an FPGA (field-programmable gate array) or an ASIC (application-specific integrated circuit). The processing unit **22** also comprises one or more memories **24** connected to or integrated in the processing component **23**. At least one of these memories **24** forms a computer-readable storage medium on which at least one computer program is stored, said computer program comprising instructions which cause the processing unit **22** to execute the steps of the current measurement method, which will be described later.

[0060] In this case, the processing unit **22** comprises a “metrology” microcontroller **23a** which in particular acquires the measurements taken by the sensors of the meter **10** and carries out certain processing operations on said measurements, and an “application” microcontroller **23b**, which in particular controls the cut-off member **18**. The current measurement method is implemented in the metrology microcontroller **23a**.

[0061] The current measurement device **20** specifically will now be discussed.

[0062] The current measurement device **20** allows a main current passing through said device to be measured. The main current in this case is the current I flowing on the phase line **14** and the phase conductor **16** and consumed by the installation **11**.

[0063] With reference to FIGS. **3** and **4**, in one embodiment the current measurement device **20** comprises two parts fastened to each other: a first part **30** and a second part **31**.

[0064] The first part **30** comprises a shunt **32**.

[0065] The shunt **32** comprises a main segment **33** in the form of a rectangular plate, and a first terminal **34a** and a second terminal **34b** located at the ends of the main segment **33**.

[0066] The two terminals **34a**, **34b** each extend from the same side of the main segment **33**, perpendicularly to the length of the main segment **33**.

[0067] The shunt **32** is made of at least one material, in this case comprising Manganin (CuMn12Ni).

[0068] The first part **30** furthermore comprises plates **35a**, **35b**, in this case made of copper, which are fastened to the ends of the shunt **32** on either side thereof. The plates **35a**, **35b** are, for example, welded to the ends of the shunt **32**. The side faces **36** of the shunt **32** and the side faces **37** of the plates **35a**, **35b** are coplanar. The underside **38** and top surface **39** of the shunt **32** are coplanar, as are the underside **40** and top surface **41** of the plates **35a**, **35b**.

[0069] Here, the term “top surface” means on the side having the terminals **34a**, **34b** of the shunt **32**, and the term “underside” means the other side.

[0070] The first part **30** thus forms a first element which is electrically conductive.

[0071] The second part **31** is generally in the shape of a rectangular plate.

[0072] The second part **31** is made of at least one material, in this case comprising a copper alloy, for example Cu-ETP C11000.

[0073] The second part **31** comprises a side **43** (its upper length in this case) in which there is formed a notch **44** that defines a recessed portion **45** extending along a segment of the length of the second part **31**, and a non-recessed portion **46** extending along said segment of the length of the second part **31**.

[0074] The non-recessed portion **46** forms a second element which is electrically conductive. The second element **46** is therefore a segment of the second part **31**.

[0075] The first element **30** extends into the notch **44** so as to leave a non-electrically conductive space **47** between the first element **30** and the second element **46**. This space **47** is empty in this case, but it could be filled with a non-conductive material.

[0076] Here, each end of the first element **30** comprises a first cavity **48** formed in its body, and each end of the second element **46** comprises a second cavity **49** formed in its body. The first element **30** and the second element **46** have their ends nested in such a way that the bottom of the first cavity **48** in each end of the first element **30** abuts the bottom of the second cavity **49** in a separate end of the second element **46**.

[0077] Each end of the first element **30** is fastened to a separate end of the second element **46** by an electrically conductive connection. In this case, each electrically conductive connection is established by an electrical spot-welding operation. Spot-welding two metal bodies involves maintaining a high pressure between the two metal bodies that are to be welded. A very high electric current is then applied for a very short period of time by means of electrodes. This electric current heats the contact surfaces, which remain fastened together as they cool.

[0078] The position of the spot welds **55a**, **55b** can be seen in FIG. 4.

[0079] It should also be noted that the top surface **50** of the first element **30** and the top surface **51** of the second part **31** are coplanar, and that the side faces **52** of the first element **30** and the side faces **53** of the second part **31** are coplanar.

[0080] Thus, when the main current I passes through the current measurement device **20**, the main current is split into a first current I_1 flowing in the first element **30** and a second I_2 flowing in the second element **46**. The current measurement device **20** therefore comprises a current splitter comprising a first branch formed by the first element **30** and a second branch formed by the second element **46**. The main current I is distributed between the two branches according to their respective impedances.

[0081] Part of the current to be measured is therefore diverted from the main copper bar, which is formed by the second part **31**, in particular by its non-recessed portion **46**, into a secondary bar in which the measurement shunt **32** is located. The secondary bar is welded over the main copper bar.

[0082] The first current I_1 is therefore representative of the main current I (it is a “resemblance” thereof).

[0083] Owing to the shunt **32**, the impedance of the first element **30** is much greater than the impedance of the second element **46**. The first current I_1 passing through the first element **30** is thus much lower than that passing through the second element **46**. The power dissipated in the shunt **32** is therefore much lower than when the entire main current passes through a shunt. Advantageously, the electrical resistance of the first element **30** is between 5 and 100 times the electrical resistance of the second element **46**.

[0084] By way of example, the shunt **32** has a length $l_{\text{sub.2}}$ of 10 mm, a height $h_{\text{sub.2}}$ of 4 mm and a thickness $e_{\text{sub.2}}$ of 1.79 mm, and therefore a cross section along a plane perpendicular to its length $l_{\text{sub.2}}$ of 7.17 mm^{sup.2}. The impedance of the shunt **32** at 20° C. is then 599.9 Ω .

[0085] The two plates **35a**, **35b** each have a length of 10 mm, a height of 4 mm and a thickness of 1.79 mm. The total impedance of the two plates at 20° C. is 48.2 $\mu\Omega$.

[0086] The cross section of the first element **30** along a plane perpendicular to its length is 7.2 mm^{sup.2}.

[0087] The total impedance of the first element **30** at 20° C. is 648.0μΩ.

[0088] The second element **46** has a length **13** of 30 mm, a height h.sub.3 of 10 mm and a thickness of 1.79 mm. The cross section of the second element **46** along a plane perpendicular to its length is 17.9 mm.sup.2.

[0089] The impedance of the second element **46** at 20° C. is 28.9μΩ.

[0090] It is therefore clear that the second element has a significantly lower resistance, which results from the difference in the resistance of the materials used (the resistance of Manganin is significantly higher than that of copper).

[0091] Thus, when a main current I of 200 A passes through the current measurement device **20**, the first current **I1** passing through the first element **30** is a current of 8.5 A, and the second current **I2** passing through the second element **46** is a current of 191.5 A.

[0092] The main current I can therefore be evaluated on the basis of the measurement of the first current **I1**, and therefore on the basis of the measurement of the voltage across the terminals of the shunt **32**.

[0093] The processing unit **22** measures the voltage across the terminals of the shunt **32** and then evaluates the main current I on the basis of said voltage. The processing unit **22** may optionally evaluate first the first current **I1** on the basis of the voltage and then evaluate the main current I on the basis of the first current **I1**.

[0094] Ideally, the distribution of currents between the two branches should always remain the same regardless of the temperature, so that the measurement remains unchanged and the main current I can be obtained by multiplying the first current **I1** (or the voltage across the terminals of the shunt **32**) by a constant coefficient.

[0095] However, as has been seen, the shunt **32** comprises Manganin and the plate **31** comprises a copper alloy. These two materials do not have the same thermal resistance. The temperature drift of the resistance of copper is about 3.93×10.sup.-3/° C., while the resistance of Manganin drifts by 0.001×10.sup.-3/° C.

[0096] Depending on the temperature, therefore, the resistance of the second element **46** varies more quickly than that of the first element **30**. The current in the shunt **32** thus varies depending on the temperature, thereby generating a measurement error.

[0097] Table 1 in the annex includes the voltage values across the terminals of the shunt **32** (in V) as a function of the temperature (between -40° C. and +70° C.), as well as the measurement error relating to the point measured at 20° C., which is the reference temperature during calibration of the meter **10**.

[0098] It can be seen that the measurement error may be relatively large.

[0099] However, this measurement error can be compensated for by using temperature measurements produced by the aforementioned temperature sensor **21**.

[0100] The processing unit **22** compensates, depending on said temperature measurements, for the temperature drift difference of the electrical resistance of the first element **30** and of the electrical resistance of the second element **46**.

[0101] The processing unit **22**: [0102] measures a voltage across the terminals of the shunt **32**; [0103] acquires temperature measurements; [0104] determines, depending on the temperature, a value of a compensation parameter for compensating for the temperature drift difference of the electrical resistance of the first element **30** and of the electrical resistance of the second element **46**; [0105] corrects the voltage across the terminals of the shunt **32** using said value of said compensation parameter; [0106] evaluates the main current I on the basis of the corrected voltage.

[0107] In one embodiment, the compensation parameter is equal to a constant voltage value per unit of temperature. By way of example, this constant value is 17.9 mV/° C.

[0108] To the voltage measurement across the terminals of the shunt, the processing unit **22** adds the value of the compensation parameter corresponding to the measured temperature.

[0109] Table 2 in the annex is thus obtained.

[0110] It can be seen that the measurement error is significantly lower, reduced to lower values than the dimensions imposed by the standards.

[0111] It would of course be possible to compensate for the temperature drift difference in a different way or by using additional operations. It would be possible, for example, to additionally add a constant positive offset to the corrected voltage of Table 2 to further reduce the measurement error.

[0112] The invention thus makes it possible to measure high currents (for example 200 A or 320 A) using a standard shunt that is normally used for currents of less than 100 A.

[0113] In the region of the heating, the power dissipated at a current of 200 A in the current measurement device **20** is 1.11 W compared with 7.2 W in a “conventional” shunt. The power dissipated is therefore significantly reduced and is even less than the power of 1.46 W dissipated in a conventional shunt at 90 A.

[0114] The cost of the current measurement device is low because it uses a shunt of known (and inexpensive) design, as well as very inexpensive additional elements. The current measurement device can therefore be integrated into a pre-existing meter that was initially designed to measure low currents, without significantly modifying said meter, to obtain one capable of measuring high currents (typically greater than 100 A). The current sensor in particular is already present in the pre-existing meter in order to compensate for the temperature drift of the shunt.

[0115] It goes without saying that the invention is not limited to the described embodiment but covers any variant falling under the scope of the invention as defined by the claims.

[0116] The first element could comprise only the shunt, without the copper plates described herein positioned on either side of the shunt. The shunt could have a different shape.

[0117] Likewise, the second part and the second element could have a different shape.

[0118] It has been described that the first element is an “individual” part whereas the second element is a segment of a part (of the second part **31**); indeed, the main current flows well in the second part on either side of the second element. This configuration is not mandatory, and the second element could also be an individual part welded to the first element. For example, the second element could comprise two arms, each fastened to one of the ends of the first element.

[0119] All the materials mentioned here could be different.

[0120] The electricity meter could be a multi-phase meter.

[0121] The apparatus in which the current measurement device is integrated need not necessarily be an electricity meter; it could be any apparatus within which current is measured.

Annexes

TABLE-US-00001 TABLE 1 Shunt Temp. [° C.] voltage [V] Error [%] -40 4.017 -21.47% -30 4.205 -17.81% -20 4.390 -14.19% 0 4.756 -7.03% 10 4.937 -3.50% 20 5.116 0.00% 30 5.293 3.47% 40 5.469 6.90% 50 5.643 10.31% 60 5.816 13.69% 70 5.987 17.04%

TABLE-US-00002 TABLE 2 Shunt Corrected Corrected Temp. [° C.] voltage [V] Error [%] voltage [V] error [%] -40 4.017 -21.47% 5.091 -0.48% -30 4.205 -17.81% 5.100 -0.32% -20 4.390 -14.19% 5.106 -0.19% 0 4.756 -7.03% 5.114 -0.03% 10 4.937 -3.50% 5.116 0.00% 20 5.116 0.00% 5.116 0.00% 30 5.293 3.47% 5.114 -0.03% 40 5.469 6.90% 5.111 -0.09% 50 5.643 10.31% 5.106 -0.19% 60 5.816 13.69% 5.100 -0.31% 70 5.987 17.04% 5.092 -0.46%

Claims

1. A current measurement device arranged to measure a main current passing therethrough and comprising: a first element which is electrically conductive and which includes a measurement shunt; a second element which is electrically conductive, the second element being a segment of a part, said part being generally in the shape of a rectangular plate; each end of the first element being fastened to a separate end of the second element by an electrically conductive connection such that the main current is split into a first current flowing in the first element and a second

current flowing in the second element, the measurement shunt thus making it possible to measure the first current, which is representative of the main current.

2. The current measurement device according to claim 1, wherein said part comprises a side in which there is formed a notch that defines a recessed portion extending along a segment of a length of the part, and a non-recessed portion extending along said segment of the length of the part, the second element being said non-recessed portion, and the first element extending in the notch so as to leave a non-electrically conductive space between the first element and the second element.

3. The current measurement device according to claim 1, wherein each end of the first element comprises a first cavity formed in its body, and each end of the second element comprises a second cavity formed in its body, the first element and the second element having their ends nested in such a way that a bottom of the first cavity in each end of the first element abuts a bottom of the second cavity in a separate end of the second element.

4. The current measurement device according to claim 1, wherein the second element is made of at least one material comprising a copper alloy.

5. The current measurement device according to claim 1, wherein the first element comprises the measurement shunt and two plates, which are fastened to the measurement shunt and positioned on either side of the measurement shunt.

6. The current measurement device according to claim 1, wherein the measurement shunt is made of at least one material comprising Manganin.

7. The current measurement device according to claim 1, wherein each electrically conductive connection is established by a spot-welding operation.

8. The current measurement device according to claim 1, wherein an electrical resistance of the first element is between 5 times and 100 times an electrical resistance of the second element.

9. An electrical apparatus including the current measurement device according to claim 1, the electrical apparatus comprising a processing unit arranged to: measure a voltage across the terminals of the measurement shunt; evaluate the main current on the basis of said voltage.

10. The electrical apparatus according to claim 9, further comprising a temperature sensor, the processing unit being further arranged to acquire temperature measurements produced by the temperature sensor and to compensate, depending on said temperature measurements, for a temperature drift difference of an electrical resistance of the first element and an electrical resistance of the second element.

11. The electrical apparatus according to claim 9, the electrical apparatus being an electricity meter.

12. The electrical apparatus according to claim 9, the electrical apparatus being arranged to measure a main current greater than 100 A using the current measurement device.

13. A current measurement method, carried out in a processing unit of the electrical apparatus according to claim 10 and comprising the steps of: measuring a voltage across the terminals of the measurement shunt; acquiring the temperature measurements produced by the temperature sensor; determining, depending on the temperature, a value of a compensation parameter for compensating for the temperature drift difference of the electrical resistance of the first element and of the electrical resistance of the second element; correcting the voltage across the terminals of the measurement shunt using said value of said compensation parameter; evaluating the main current on the basis of the corrected voltage.

14. (canceled)

15. A non-transitory computer-readable storage medium on which a computer program is stored, wherein the computer program comprises instructions that cause a processing unit of an electrical apparatus to execute the steps of the current measurement method according to claim 13.
